

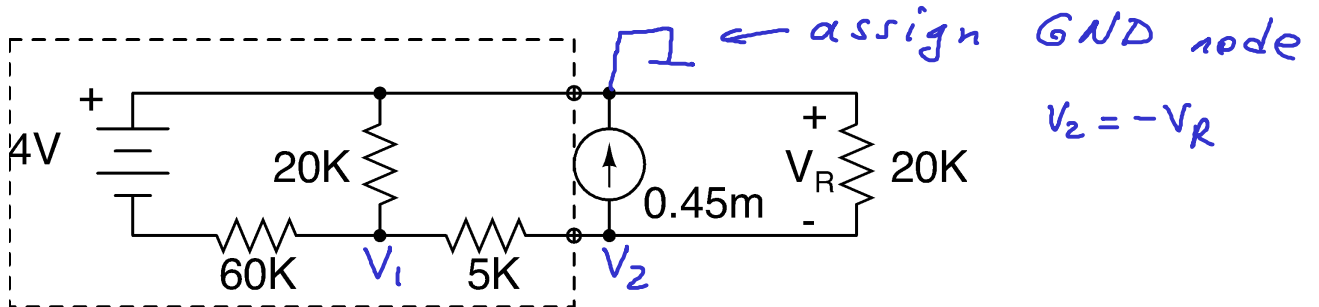
MIDTERM EXAM SOLUTIONS

Question 1 [25 Points]:

For the circuit below, find the voltage V_R using the following two methods:

(a) By using nodal analysis.

(b) By simplifying the circuit in the dashed box using Thévenin and/or Norton equivalent circuits.



a) $V_1: \frac{V_1 - (-4)}{60} + \frac{V_1}{20} + \frac{V_1 - V_2}{5} = 0 \Rightarrow V_1 \left(\frac{1}{60} + \frac{1}{20} + \frac{1}{5} \right) = \frac{V_2}{5} - \frac{4}{60}$

$V_2: \frac{V_2 - V_1}{5} + 0.45 + \frac{V_2}{20} = 0 \Rightarrow V_1 \left(\frac{16}{60} \right) = \frac{V_2}{5} - \frac{4}{60}$

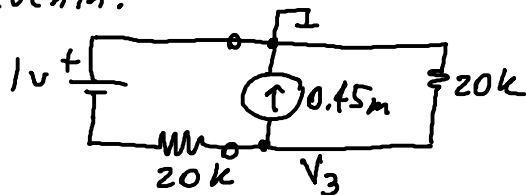
$\Rightarrow V_1 = \frac{3}{4} V_2 - \frac{1}{4}$

$V_2 \left(\frac{1}{5} - \frac{1}{5} \cdot \frac{3}{4} + \frac{1}{20} \right) = -0.45 - \frac{1}{5} \cdot \frac{1}{4} \Rightarrow V_2 = -5 \text{ V}$

$V_R = 5 \text{ V}$

b) $R_{eq} = 60 \parallel 20 + 5 = 20 \text{ K}$

Thévenin:



$V_{TH} = \frac{20}{60+20} 4 = 1 \text{ V}$

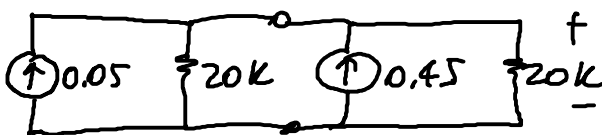
$\frac{V_3 - (-1)}{20} + 0.45 + \frac{V_3}{20} = 0$

$V_3 \left(\frac{1}{20} + \frac{1}{20} \right) = -0.45 - \frac{1}{20} = -0.5$

$V_3 = -5 \text{ V}$ $V_R = 5 \text{ V}$

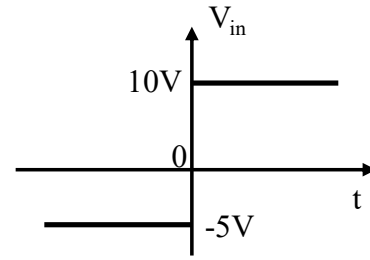
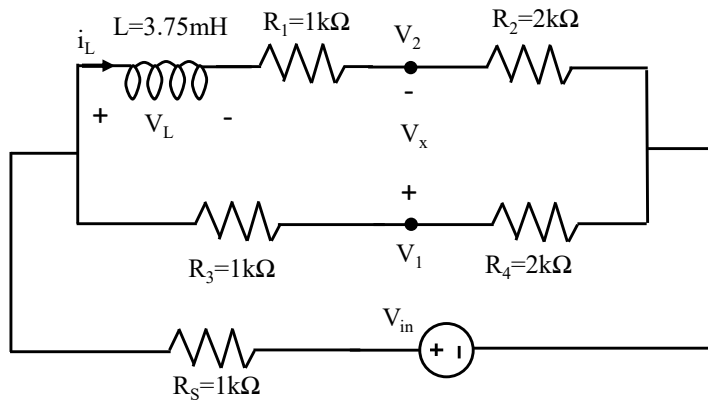
Norton: $I_N = \frac{4}{5 \parallel 20 + 60} \cdot \frac{20}{20+5} = 0.05 \text{ mA}$

current divider



$V_R = (0.45 + 0.05)(20 \parallel 20) = 5 \text{ V}$

Question 2 [25 points]:



For the circuit above,

(a) Find the value of $V_x(t)$ as $t \rightarrow \infty$.

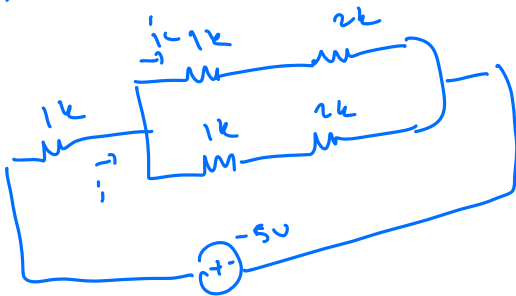
(b) Find and plot $i_L(t)$ for all t .

(c) Find and plot $V_x(t)$ for all t . Note that $V_x(t) = V_1(t) - V_2(t)$.

a.) in steady state $L \rightarrow$ short ckt. $\frac{R_1}{R_2} = \frac{R_3}{R_4}$ so $V_1 = V_2$ and $V = 0$

b) for $i_L(0)$ $V_{in} = -5V$ for a long time.

$$i = \frac{-5}{1+1.5} = -2mA \quad i_L(0) = -1mA$$



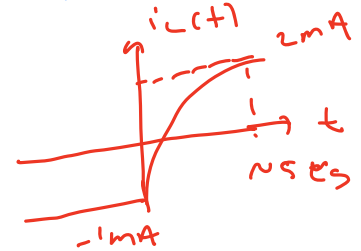
Similarly for $i_L(\infty)$ $V_{in} = 10V$ $i_L(\infty) = 2mA$

$$i_L(t) = i_f + (i_i - i_f) e^{-t/\tau}$$

$$i_L(t) = 2 + (-1-2) e^{-t/12} = 2 - 3e^{-t/12} \text{ mA}$$

$$R_{eq} = 3 + (3||1) = 3 + \frac{3}{4} = 3.75k\Omega$$

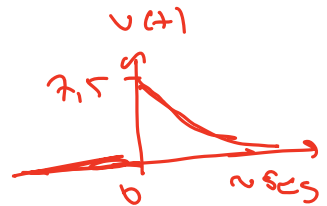
$$\tau = \frac{3.75mH}{3.75k\Omega} = 1\mu s$$



$$c) \quad V_2 = R_2 (i_L(t)) = 2 \cdot (2 - 3e^{-t/12}) = 4 - 6e^{-t/12}$$

$$V_1 = \frac{R_4}{R_3 + R_4} [V_L + i_L(R_1 + R_2)]$$

$$V_L = L \frac{di_L}{dt} = L \cdot \frac{3}{2} e^{-t/12} = 3R_{eq} \cdot e^{-t/12} = 3 \cdot 3.75 \cdot e^{-t/12} \text{ V}$$



$$V_1 = \frac{2}{3} [3 \cdot 3.75 \cdot e^{-t/12} + (2 - 3e^{-t/12}) 3] = 7.5e^{-t/12} + 4 - 6e^{-t/12} = 4 + 1.5e^{-t/12}$$

$$V(t) = V_1(t) - V_2(t) = 4 + 1.5e^{-t/12} - 4 + 6e^{-t/12} = 7.5e^{-t/12} \text{ V}$$

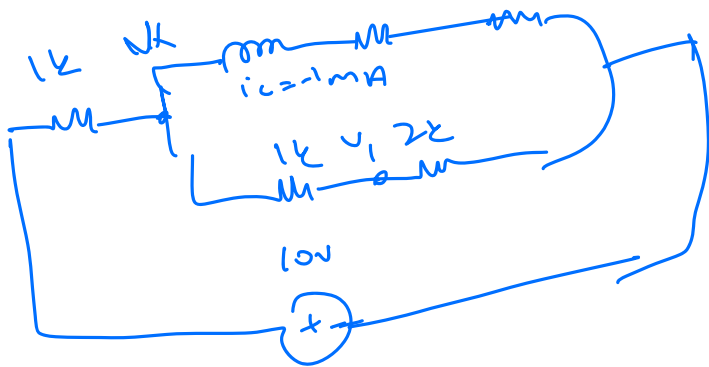
$V(t)$ should be zero for $t < 0$, jumps @ $t=0$ and eventually goes to zero.

the other solution is calculation $v_1(t)$ and $v_2(t)$ w/o $v_L(t)$

$$V_2(t) = i_L(t) R_2 = 4 - 6e^{-t/2}$$

for $v_1(t)$ $V_f = 2 \cdot 2 \text{ mA} = 4 \text{ V}$

for v_f



$$\frac{v_x - 10}{1} + \frac{v_x}{3} = 0$$

$$3v_x - 30 + v_x = 0$$

$$4v_x = 30$$

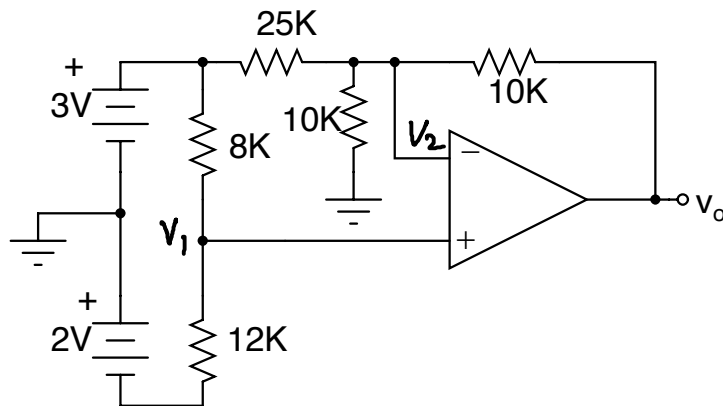
$$v_x = \frac{30}{4}$$

$$v_1 = \frac{2}{3} \cdot \frac{30}{4} = 5 \text{ V}$$

$$V(t) = 4 + 1.5e^{-t/2} - (4 - 6e^{-t/2}) = 7.5e^{-t/2} \text{ V as before}$$

Question 3 [25 points]:

For the circuit below, assume that the OPAMP is ideal and not saturated. Find the voltage v_o .



$$v_1 = v_2$$

$$v_1: \frac{v_1 - 3}{8} + \frac{v_1 - (-2)}{12} = 0$$

$$v_1 \left(\frac{1}{8} + \frac{1}{12} \right) = \frac{3}{8} - \frac{2}{12}$$

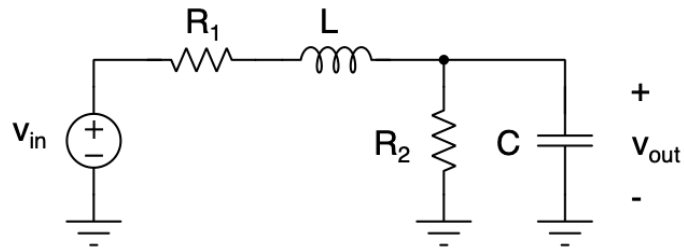
$$v_1 = \frac{24}{5} \cdot \frac{5}{24} = \frac{5}{5} = 1 = v_2$$

$$v_2: \frac{v_2 - 3}{25} + \frac{v_2}{10} + \frac{v_2 - v_o}{10} = 0$$

$$\frac{v_o}{10} = v_2 \left(\frac{1}{25} + \frac{1}{10} + \frac{1}{10} \right) - \frac{3}{25} = v_2 \left(\frac{12}{50} \right) - \frac{3}{25}$$

$$v_o = v_2 \frac{12}{5} - \frac{30}{25} = \frac{12}{5} - \frac{30}{25} = \frac{30}{25} = \underline{1.2 \text{ V}}$$

Question 4 [25 points]:



For the circuit above:

(a) Find the transfer function $H(\omega) = \frac{V_{out}}{V_{in}}$ in parametric form (i.e., in terms of R_1 , R_2 , L , C). Do not insert component values.

(b) Now, assume that $R_1 = 1 \text{ k}\Omega$, $R_2 = 3 \text{ k}\Omega$, $L = 1 \text{ }\mu\text{H}$, and $C = 1 \text{ pF}$. Find $v_{out}(t)$ if $v_{in}(t) = 5 + 2 \cos(10^9 t + \pi/3) \text{ V}$ for all t .

$$\begin{aligned} a) \quad H(\omega) &= \frac{R_2 \parallel \frac{1}{j\omega C}}{R_1 + j\omega L + (R_2 \parallel \frac{1}{j\omega C})} = \frac{\frac{R_2}{1 + j\omega R_2 C}}{R_1 + j\omega L + \frac{R_2}{1 + j\omega R_2 C}} \\ &= \frac{R_2}{(R_1 + j\omega L)(1 + j\omega R_2 C) + R_2} = \frac{R_2}{R_1 + R_2 - \omega^2 R_2 L C + j\omega(L + R_1 R_2 C)} \end{aligned}$$

b) Use superposition.

$$* \quad V_{in,1}(t) = 5 \text{ V} \Rightarrow \omega_1 = 0 \text{ rps}, \quad V_{in,1} = 5 \text{ V}$$

$$H(0) = \frac{R_2}{R_1 + R_2} = \frac{3 \text{ k}}{1 \text{ k} + 3 \text{ k}} = \frac{3}{4}$$

$$V_{out,1} = H(0) \cdot V_{in,1} = \frac{3}{4} \cdot 5 = \frac{15}{4} \text{ V} \Rightarrow \boxed{V_{out,1}(t) = \frac{15}{4} \text{ V}}$$

$$* \quad V_{in,2}(t) = 2 \cos(10^9 t + \pi/3) \text{ V} \Rightarrow \omega_2 = 10^9 \text{ rps}, \quad V_{in,2} = 2 e^{j\pi/3} \text{ V}$$

$$\begin{aligned} H(10^9) &= \frac{3 \cdot 10^3}{10^3 + 3 \cdot 10^3 - 10^{18} \cdot 3 \cdot 10^3 \cdot 10^{-12} + j \cdot 10^9 \cdot (10^{-6} + 10^3 \cdot 3 \cdot 10^3 \cdot 10^{-12})} \\ &= \frac{3 \cdot 10^3}{10^3 + j4 \cdot 10^3} = \frac{3}{1 + j4} = \frac{3}{\sqrt{17}} e^{-j \tan^{-1}(4)} \end{aligned}$$

$$V_{out,2} = H(10^9) \cdot V_{in,2} = \frac{3}{\sqrt{17}} e^{-j \tan^{-1}(4)} \cdot 2 e^{j\pi/3} = \frac{6}{\sqrt{17}} e^{j(\pi/3 - \tan^{-1}(4))}$$

$$V_{out,2}(t) = \frac{6}{\sqrt{17}} \cos(10^9 t + \pi/3 - \tan^{-1}(4))$$

$$= \boxed{1.46 \cos(10^9 t - 0.28) \text{ V}}$$

* Finally,

$$V_{out}(t) = V_{out,1}(t) + V_{out,2}(t) = \boxed{3.75 + 1.46 \cos(10^9 t - 0.28) \text{ V}}$$